Project Final Report

Date	22 February 2026
Team ID	LTVIP2026TMIDS81735
Project Name	Strategic Product Positioning Analysis
Maximum Marks	5 Marks

1.INTRODUCTION

Project Overview

This project focuses on analyzing retail sales and product placement data to understand how positioning strategies impact sales performance and consumer behavior.

• I used a large retail dataset containing product sales records, shelf placement details, and category information to identify key factors influencing revenue growth.

• The technology used in this project includes MySQL for storing and managing retail data, Tableau for creating interactive visualizations, and Flask for building the web application interface.

• The final result of the project includes 10 interactive charts and 2 detailed dashboards that allow users to analyze product performance, shelf effectiveness, and category-wise sales trends easily.

Purpose of the Project

This section explains why I created this project.

• **Business Insight:** The main goal is to help retail managers understand complex sales and placement data quickly. For example, how shelf zones, product categories, and visibility together impact overall revenue.

• **Revenue Optimization:** By creating performance indicators and placement analysis metrics, the project not only shows past sales data but also provides actionable insights to improve product positioning strategies.

• **Easy Access:** I designed a simple and user-friendly web interface so that even non-technical users can filter data, view dashboards, and understand sales insights easily.

2. IDEATION PHASE

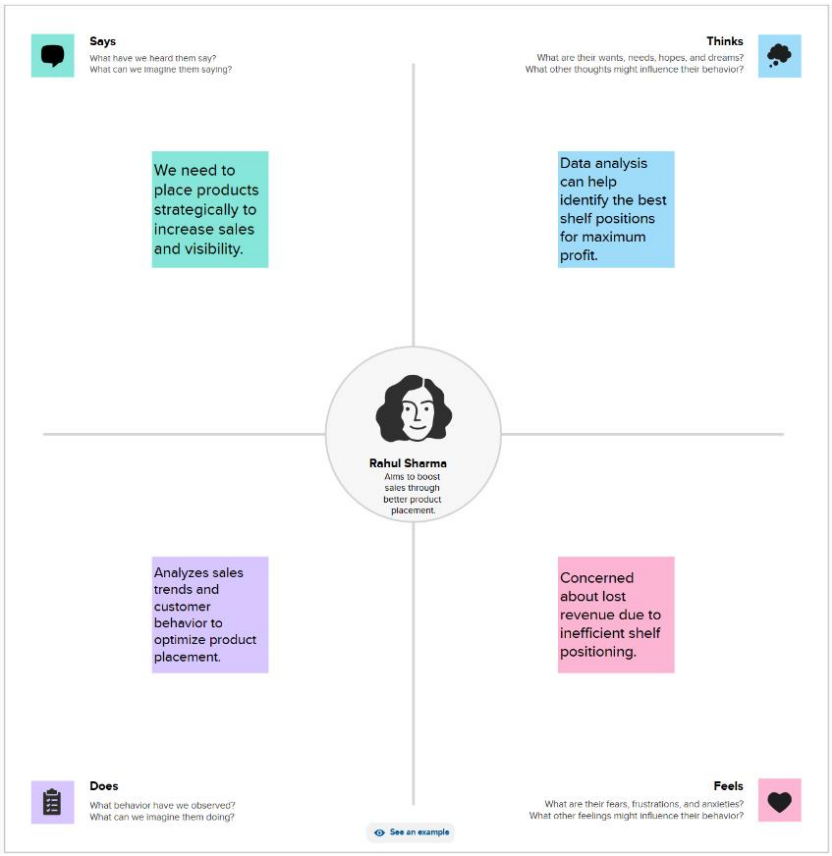
1.1 Problem Statement

1.2 Empathy map

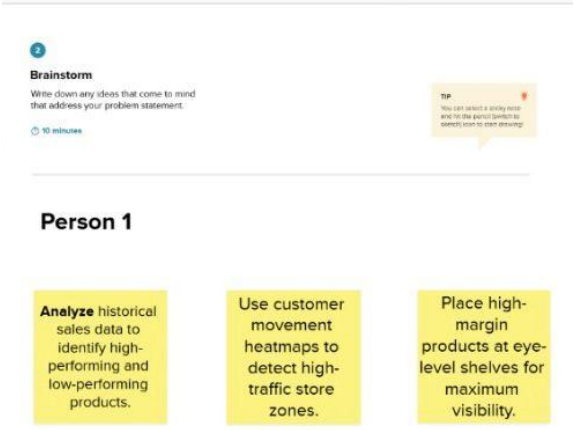
1.3 Brainstorming

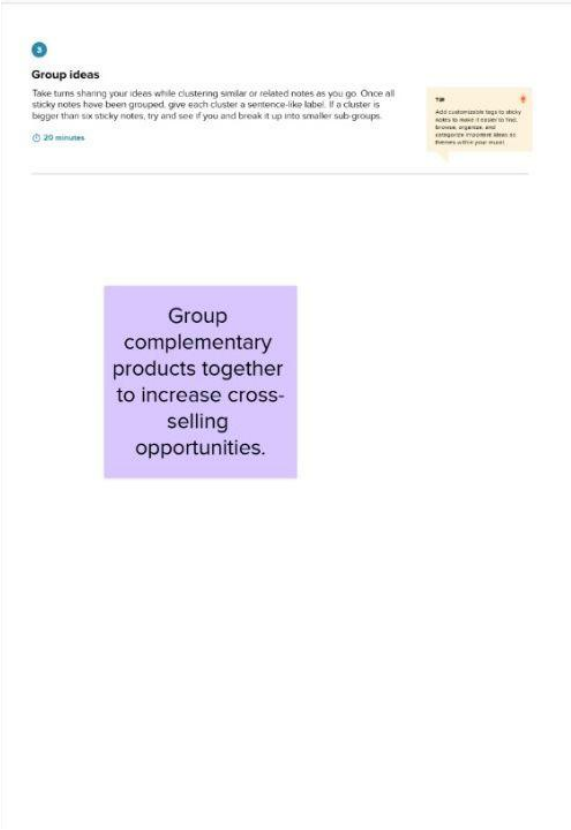
Problem Statement:

Empathy map:



Brainstorming:





4

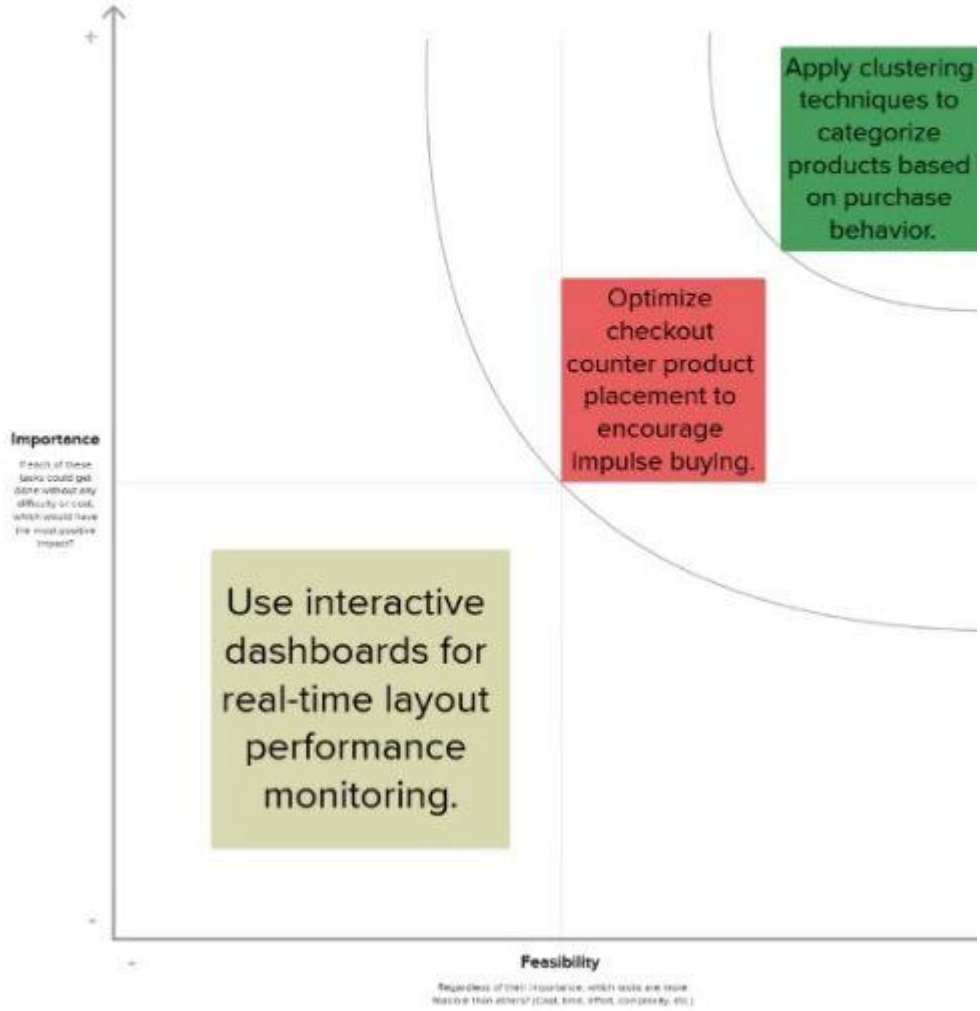
Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

20 minutes

TIP

Participants can use their cursors to point at where sticky notes should go on the grid. The facilitator can confirm the spot by using the laser pointer holding the **H key** on the keyboard.



2. REQUIREMENT ANALYSIS

2.1 Customer Journey map

2.2 Solution Requirement

2.3 Data Flow Diagram

2.4 Technology Stack

Customer Journey Map:

Scenario: [Existing experience through a product or service]	Entice How does someone become aware of this service?	Enter What do people experience as they begin the process?	Engage In the core moments in the process, what happens?	Exit What do people typically experience as the process finishes?	Extend What happens after the experience is over?
Experience steps What does the person (or people) at the center of this scenario typically experience in each step?	Review store sales performance	Realizes some products are rarely noticed	Analyze customer movement	Implement new layout	Refine placement strategy
Interactions What interactions do they have at each step along the way? - **People:** Who do they see or talk to? - **Places:** Where are they? - **Things:** What digital touchpoints or physical objects do they use?	Sales reports and dashboards	Notices product visibility gaps	Understanding the customer	Observes customer reactions	Recognizes improvement areas
Goals & motivations At each step, what is a person's primary goal or motivation? ("Help me..." or "Help me avoid...")	Help me recognize missed sales opportunities.	Help me reduce sales loss	Help me boost cross-selling	Help me monitor impact quickly	Help me adapt to seasonal trends
Positive moments What steps does a typical person find enjoyable, productive, fun, motivating, delightful, or exciting?	Clear performance insights from reports.	Easy product comparison	Logical grouping ideas	Improved product visibility	Efficient shelf utilization
Negative moments What steps does a typical person find frustrating, confusing, engering, costly, or time-consuming?	Poor visibility of high-margin products.	Unclear decision direction	Staff coordination challenges	Customer confusion initially	Seasonal demand shifts
Areas of opportunity How might we make each step better? What ideas do we have? What have others suggested?	Improve visibility of high-margin products.	Introduce visual heatmaps	Automated grouping suggestions	Customer feedback integration	Real-time customer analytics

Functional Requirements:

Following are the functional requirements of the proposed solution.

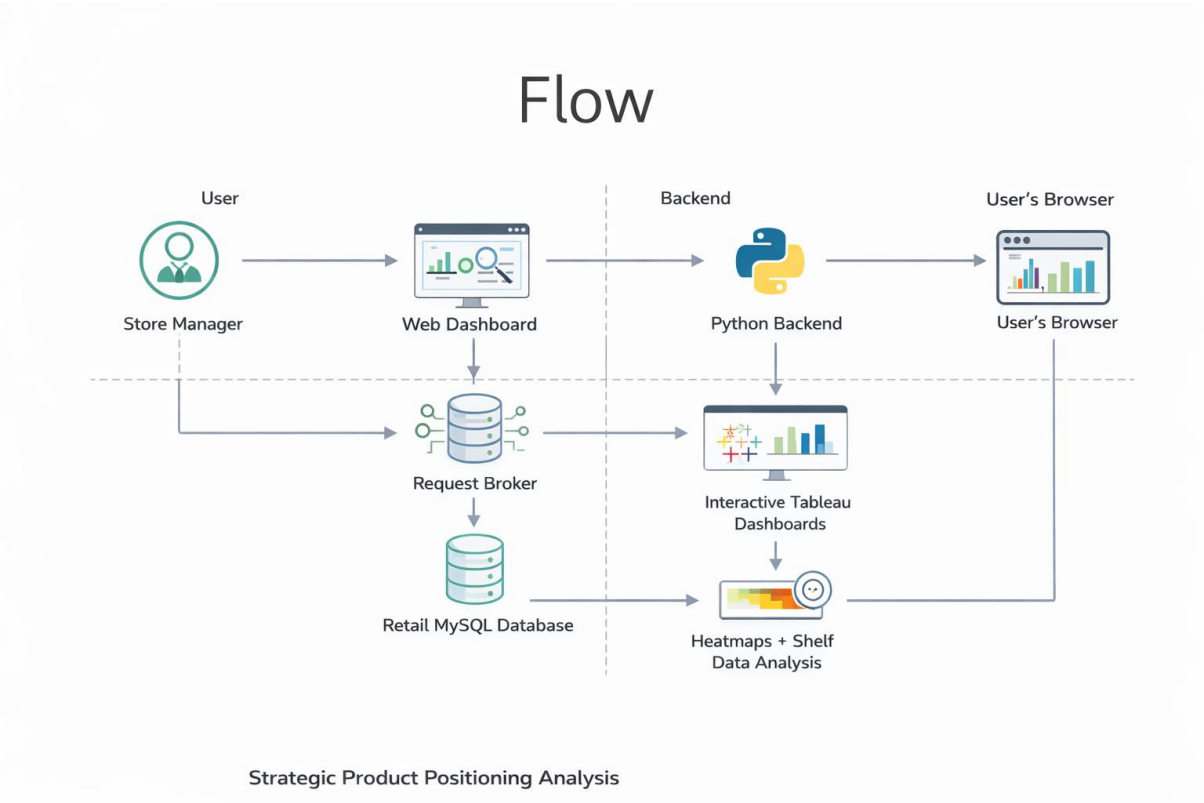
FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection	Extract product sales, profit margin, and shelf location data from the retail database using SQL queries.
FR-2	Data Visualization	Create interactive Tableau dashboards with 10+ charts showing sales trends, product performance, and shelf impact.
FR-3	Placement Analysis	Analyze product visibility, high-traffic zones, and cross-selling opportunities using heatmaps and trend analysis.
FR-4	Strategy Optimization	Provide recommendations for optimal product positioning based on sales performance and margin contribution.

Non-functional Requirements:

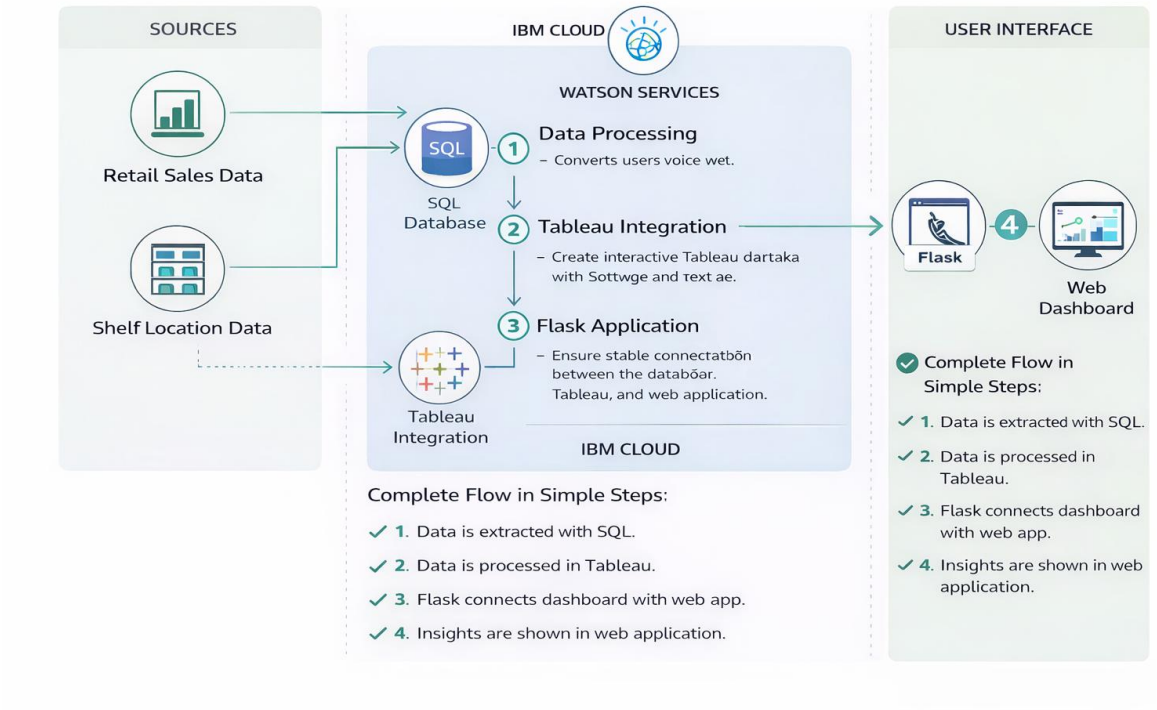
Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Provide a simple and intuitive dashboard interface for store managers.
NFR-2	Security	Protect sales and inventory data stored in the database system.
NFR-3	Reliability	Ensure stable connection between the database, Tableau, and web application.
NFR-4	Performance	Dashboards and reports must load within 3–5 seconds.
NFR-5	Availability	System should be accessible during store operational hours.
NFR-6	Scalability	Handle increasing product categories and larger sales datasets efficiently.

Data Flow Diagram:



Technology Stack:



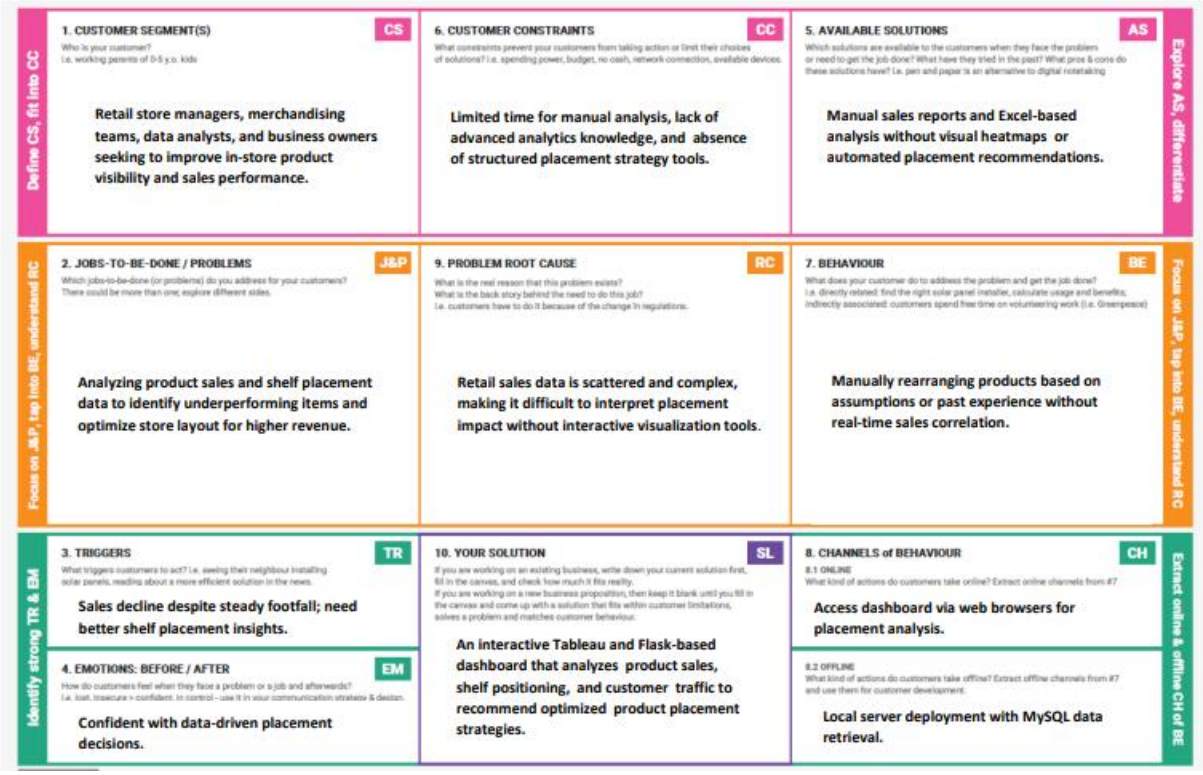
3.PROJECT DESIGN

3.1 Problem Solution Fit

3.2 Proposed Solution

3.3 Solution Architecture

Problem Solution Fit:



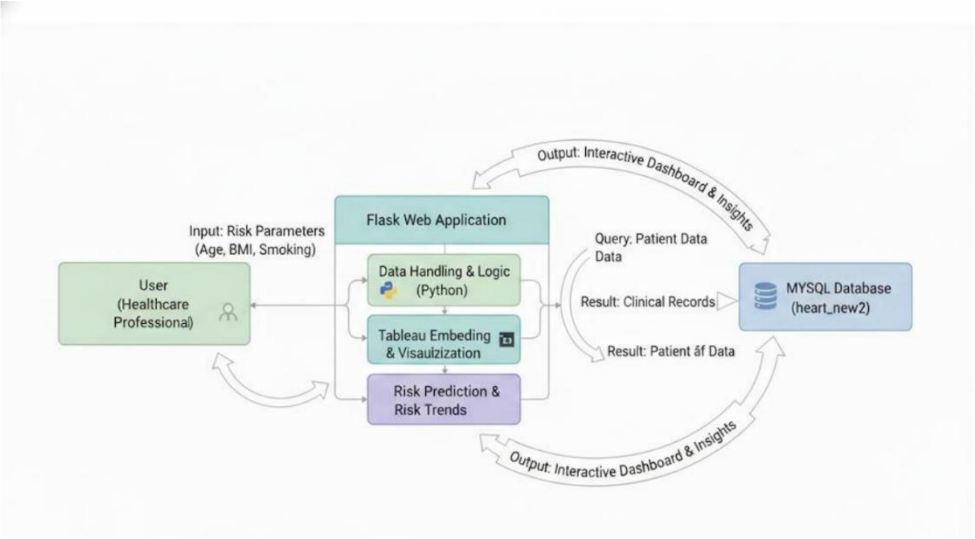
Proposed Solution:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Retailers cannot clearly identify how shelf placement affects product sales. Manual reports do not provide clear insights.
2.	Idea / Solution description	Build a Tableau and Flask dashboard to analyze sales and shelf data and suggest better product placement.
3.	Novelty / Uniqueness	Combines sales data and placement analysis in one interactive dashboard.
4.	Social Impact / Customer Satisfaction	Improves store layout and enhances customer shopping experience.
5.	Business Model (Revenue Model)	Subscription-based dashboard service for retailers.
6.	Scalability of the Solution	Can be expanded to multiple stores and large datasets.

Solution Architecture:

Solution Architecture:

Heart Disease Analysis



4.PROJECT PLANNING & SCHEDULING

4.1 Project Planning

4.2 Project Milestones & Tasks

4.3 Sprint Delivery Plan

4.4 Project Progress Tracking

Project Planning:

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	Collect retail sales and shelf position dataset.	3	High	Munazar
Sprint-1	Database Integration	USN-2	Import dataset into MySQL and connect to Tableau.	2	High	Munazar
Sprint-1	Visualization	USN-3	Build 8+ Product Positioning Visualizations.	3	Low	Munazar
Sprint-2	Analysis	USN-4	Create interactive Product Placement Dashboard.	5	Medium	Munazar
Sprint-2	Integration	USN-5	Integrate Tableau dashboard with Flask web app.	5	High	Munazar
Sprint-3	Project Admin	USN-6	Complete Phase templates and documentation.	3	Medium	Munazar
Sprint-3	Project Admin	USN-7	Final report and PDF submission.	3	Medium	Munazar
Sprint-3	Project Admin	USN-8	Record and edit Project Demo Video.	2	High	Munazar

Project Tracker, Velocity:

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	8	5 Days	04 Feb 2026	08 Feb 2026	8	08 Feb 2026
Sprint-2	10	5 Days	09 Feb 2026	13 Feb 2026	10	13 Feb 2026
Sprint-3	8	5 Days	14 Feb 2026	20 Feb 2026	8	20 Feb 2026

Velocity:

Total Story Points / Sprint Duration = 8 / 5 = 1.6

5. FUNCTIONAL AND PERFORMANCE TESTING

5.1 Performance Testing

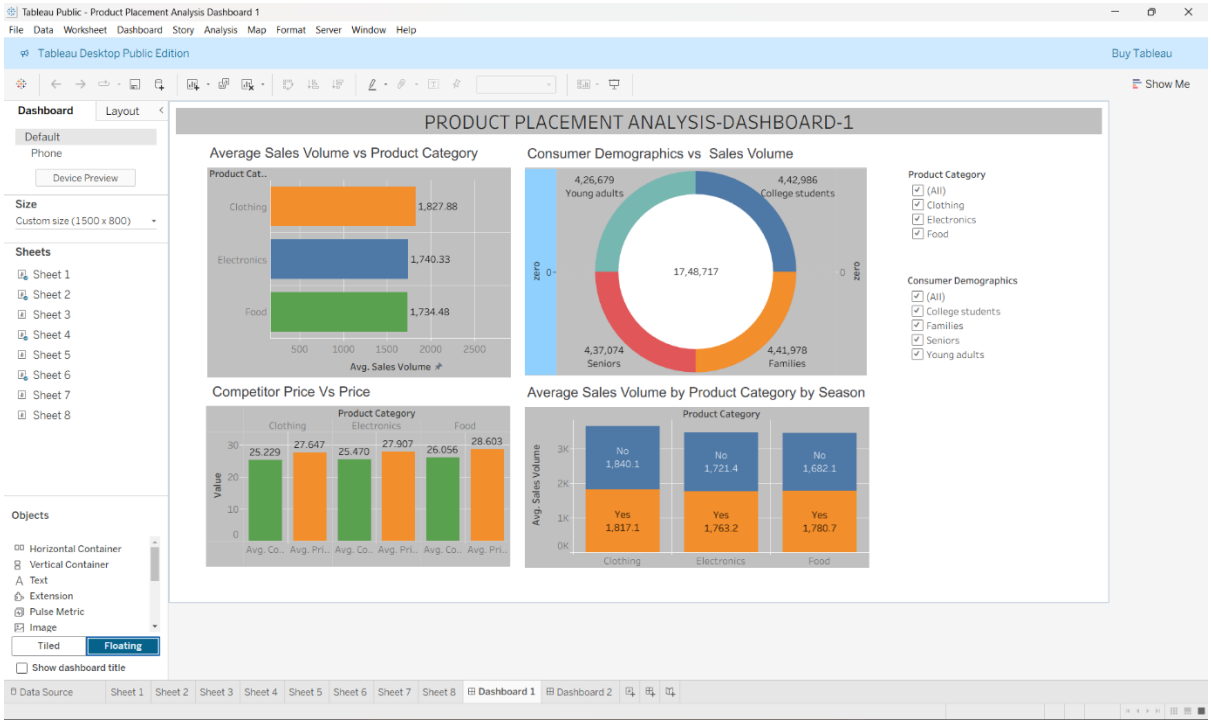
Performance Testing:

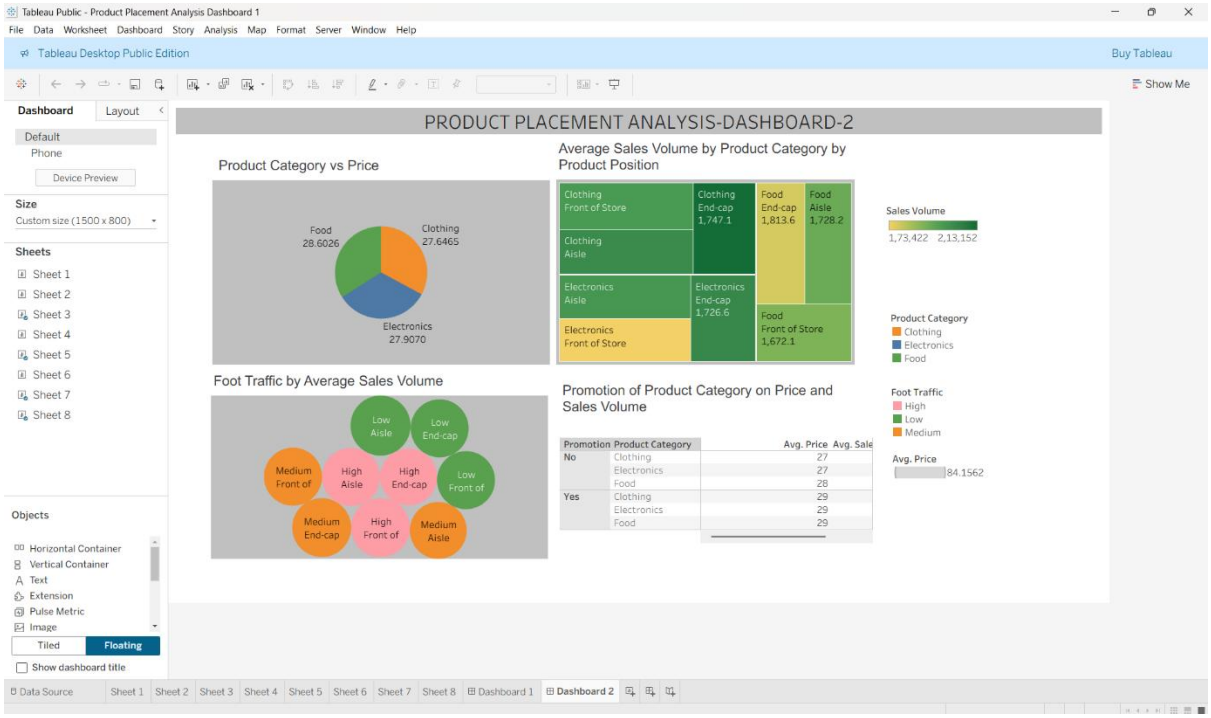
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	I cleaned the dataset using Excel/Google Sheets before importing it into Tableau. I removed duplicate records, handled missing values, corrected inconsistent entries, and standardized column names. I ensured correct data types for Price, Competitor Price, and Sales Volume before connecting the cleaned dataset to Tableau for visualization.
2.	Data Preprocessing	I removed duplicate records and handled missing values to maintain data accuracy. I assigned proper data types such as Price, Competitor Price, and Sales Volume as Measures, and Product Category and Product Position as Dimensions. Column names were standardized for clarity.
3.	Utilization of Filters	I implemented interactive filters including Product Category, Season, Product Position, Promotion Status, and Consumer Demographics. These filters dynamically update all visualizations, enabling detailed placement and sales analysis.
4.	Calculation fields Used	I created calculation fields such as Price, Competitor Price, and Sales Volume to analyze pricing impact and demand trends. These fields help compare competitor pricing and measure total product performance.
5.	Dashboard design	I developed 2 interactive dashboards integrating 8 visualizations: **1.1 Avg Sales Volume vs Product Category** **1.2 Competitor Price vs Price** **1.3 Avg Sales Volume by Product Category by Product Position** **1.4 Consumer Demographics vs Sales Volume** **1.5 Product Category vs Price** **1.6 Avg Sales Volume by Product Category by Season** **1.7 Foot Traffic by Avg Sales Volume** **1.8 Promotion of Product Category on Price and Sales Volume.**

		These dashboards provide a comprehensive view of how positioning, pricing, promotion, and demographics impact sales performance.
6	Story Design	I structured the analysis into 3 stories: 1. Category clothing has highest Average sales volume 2. The Product position at the front of the store as High sales volume and high foot traffic 3. The Product Category clothing at the front of the store ha highest avg sales. The stories present a logical flow from category performance to positioning impact.

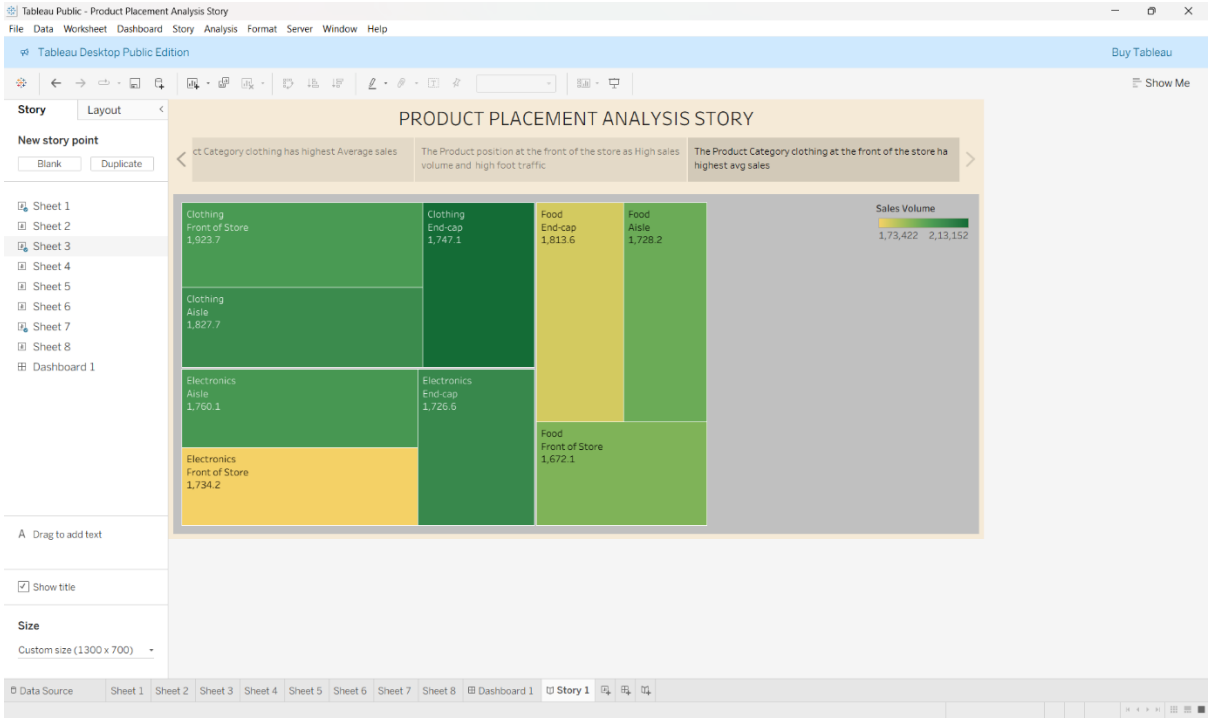
6. RESULTS

Dashboard Screenshots:

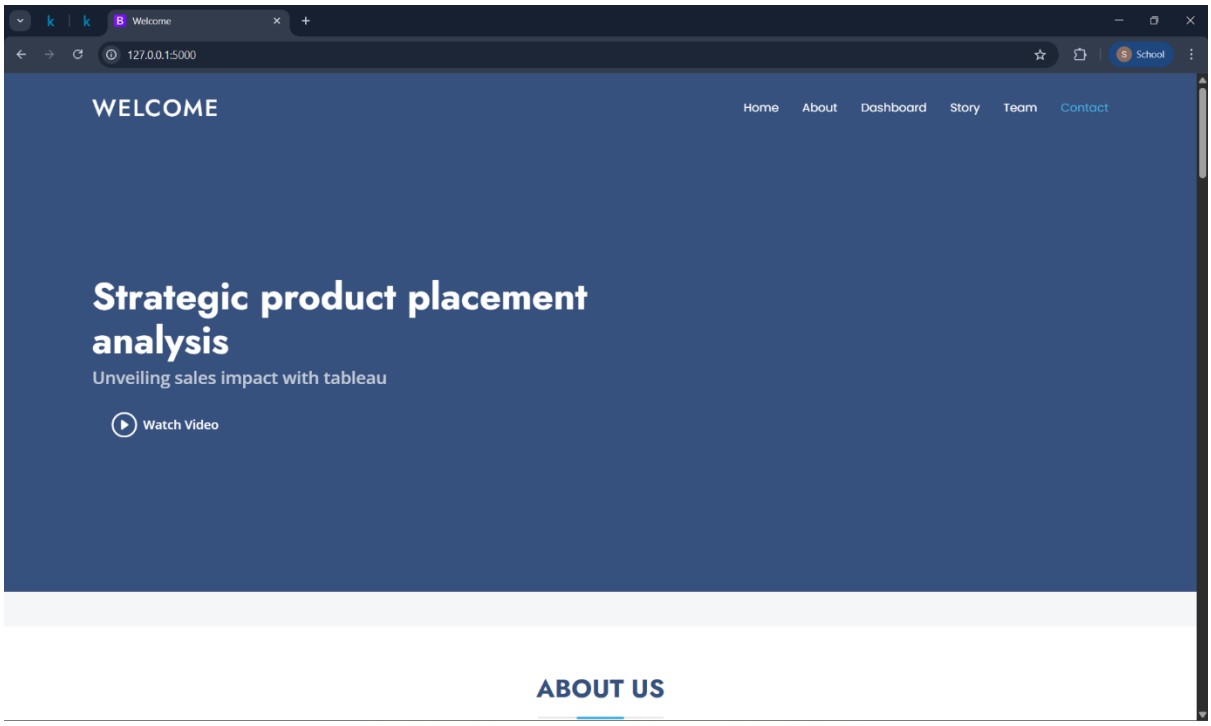


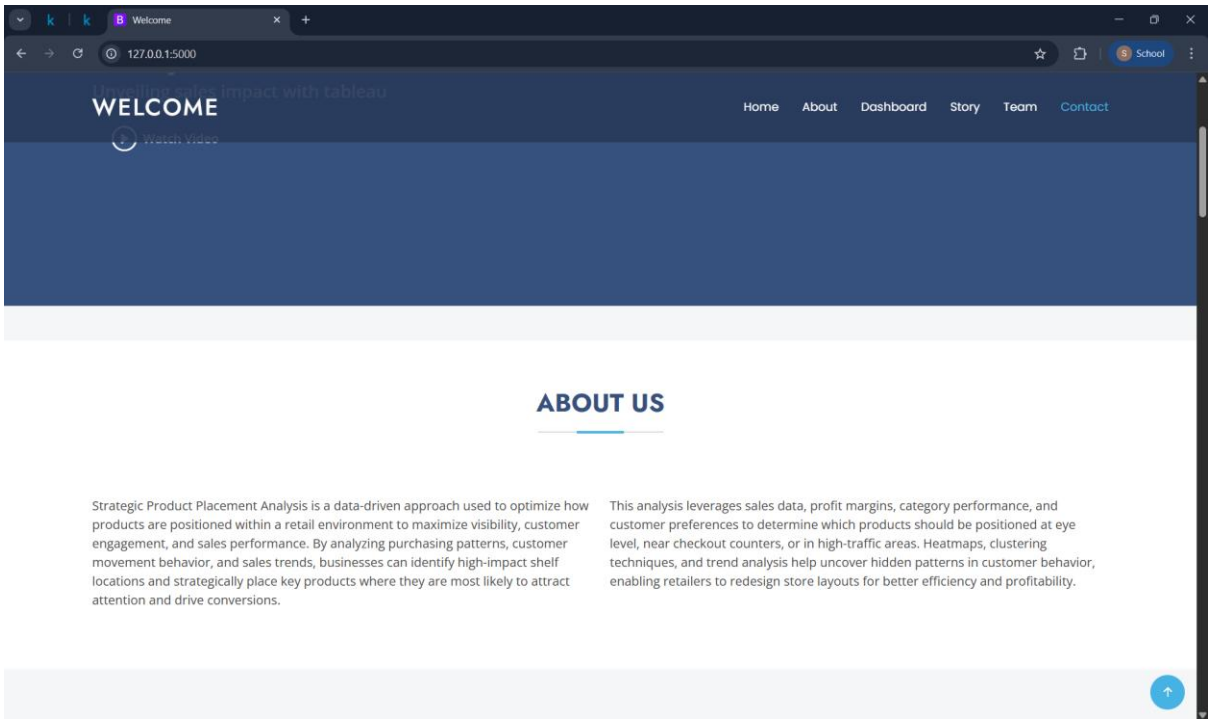


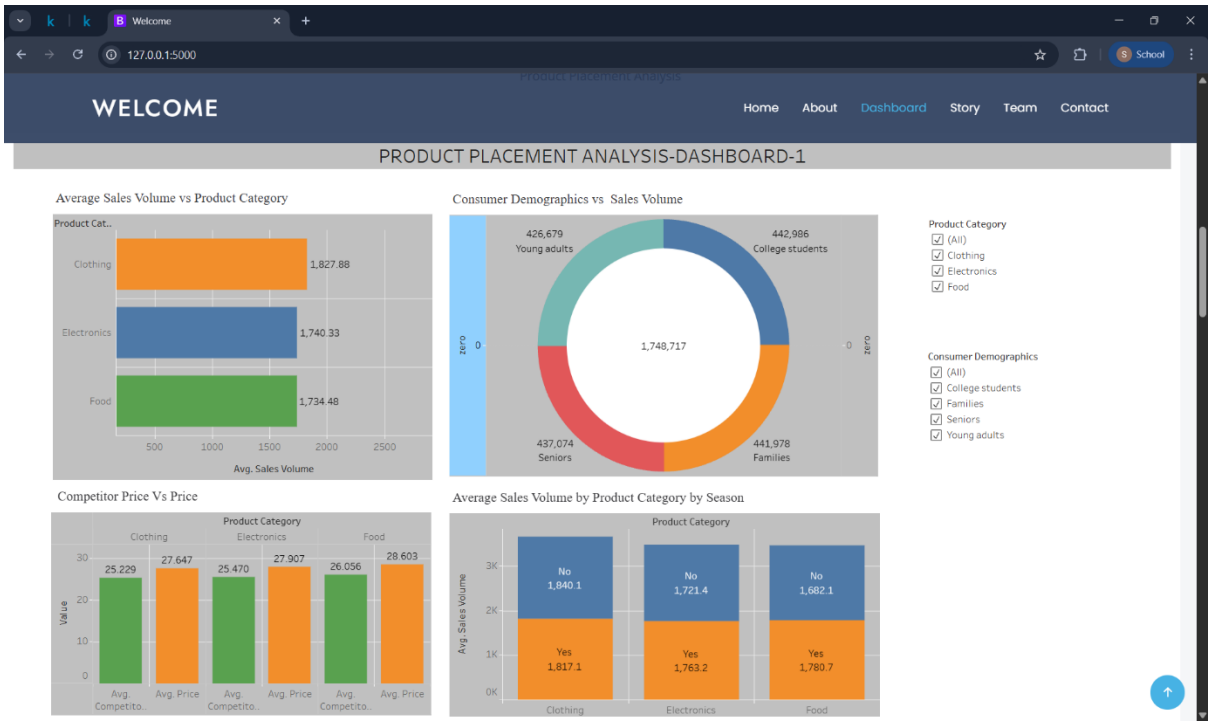
Story Screenshot:

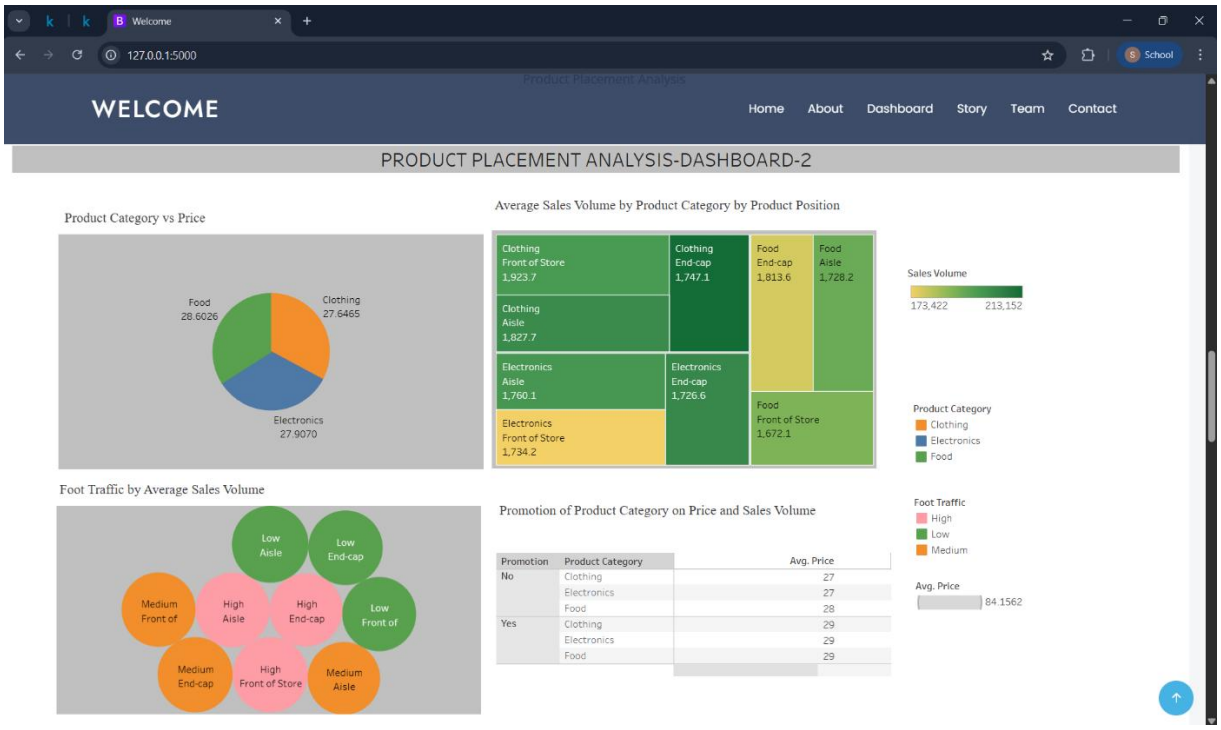


Web Integration Screenshots:

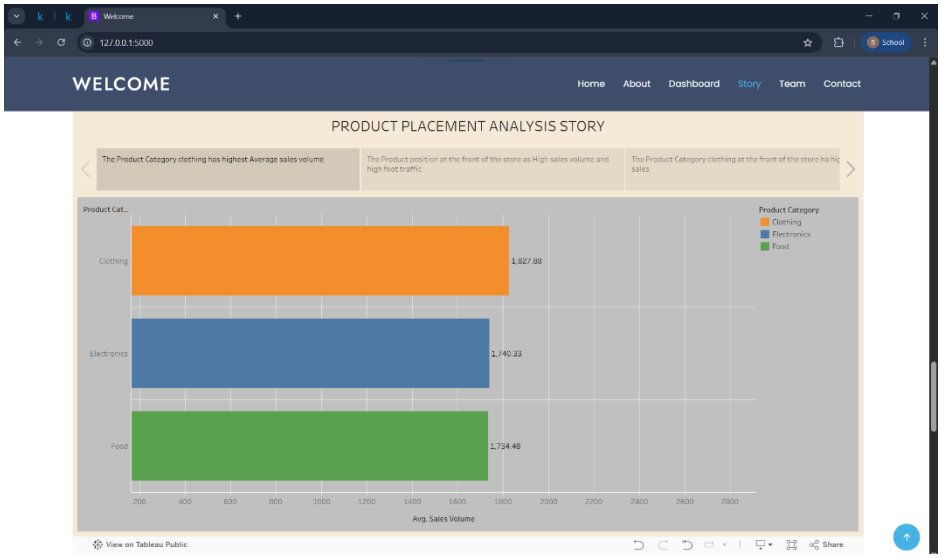


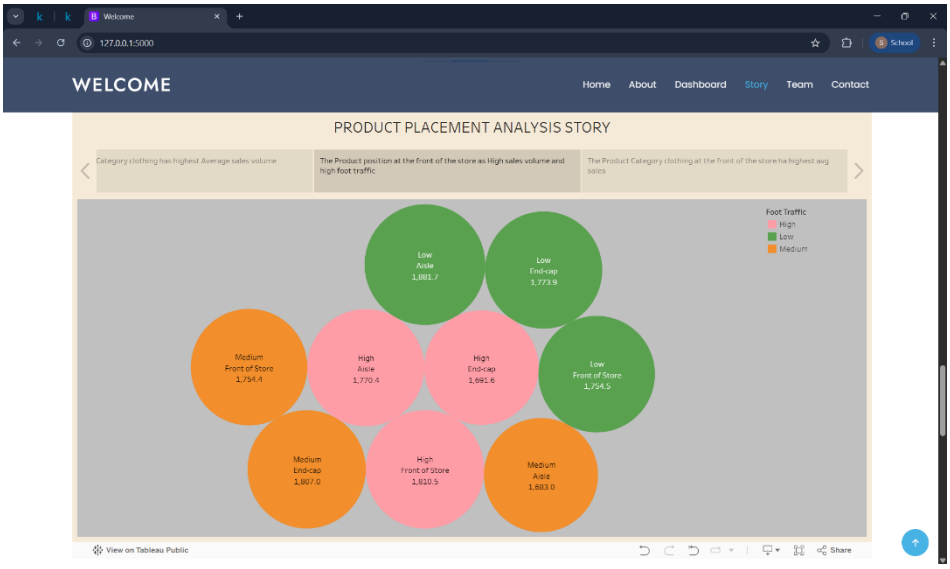






Promotion	Product Category	Avg. Price
No	Clothing	27
	Electronics	27
	Food	28
Yes	Clothing	29
	Electronics	29
	Food	29


Welcome
127.0.0.1:5000
WELCOME
Home
About
Dashboard
Story
Team
Contact
PRODUCT PLACEMENT ANALYSIS STORY
The Product Category clothing has highest Average sales volume
The Product position at the front of the store as High sales volume and high foot traffic
Product Cat..
Product Category
Clothing
Electronics
Food
1,827.88
1,740.33
1,734.48
Avg. Sales Volume
View on Tableau Public
Share


Welcome
127.0.0.1:5000
WELCOME
Home
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Story
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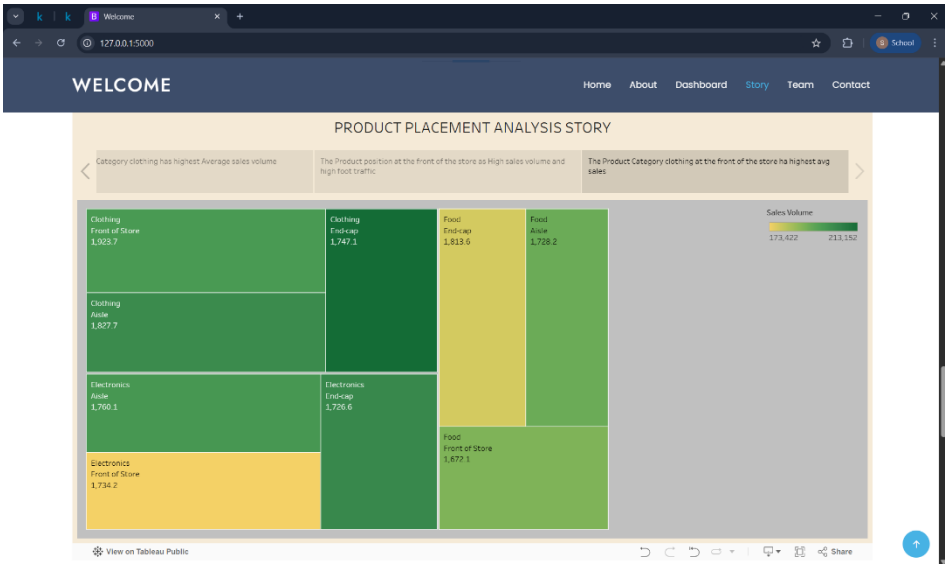
Contact
PRODUCT PLACEMENT ANALYSIS STORY
Category clothing has highest Average sales volume
The Product position at the front of the store as High sales volume and high foot traffic
The Product Category clothing at the front of the store ha highest avg sales
Foot Traffic
High
Low
Medium
Low Aisle 1,881.7
Low End-cap 1,773.9
Medium Front of Store 1,754.4
High Aisle 1,770.4
High End-cap 1,691.6
Low Front of Store 1,754.5
Medium End-cap 1,807.0
High Front of Store 1,810.5
Medium Aisle 1,683.0
View on Tableau Public
Share


Welcome
127.0.0.1:5000
WELCOME
Home
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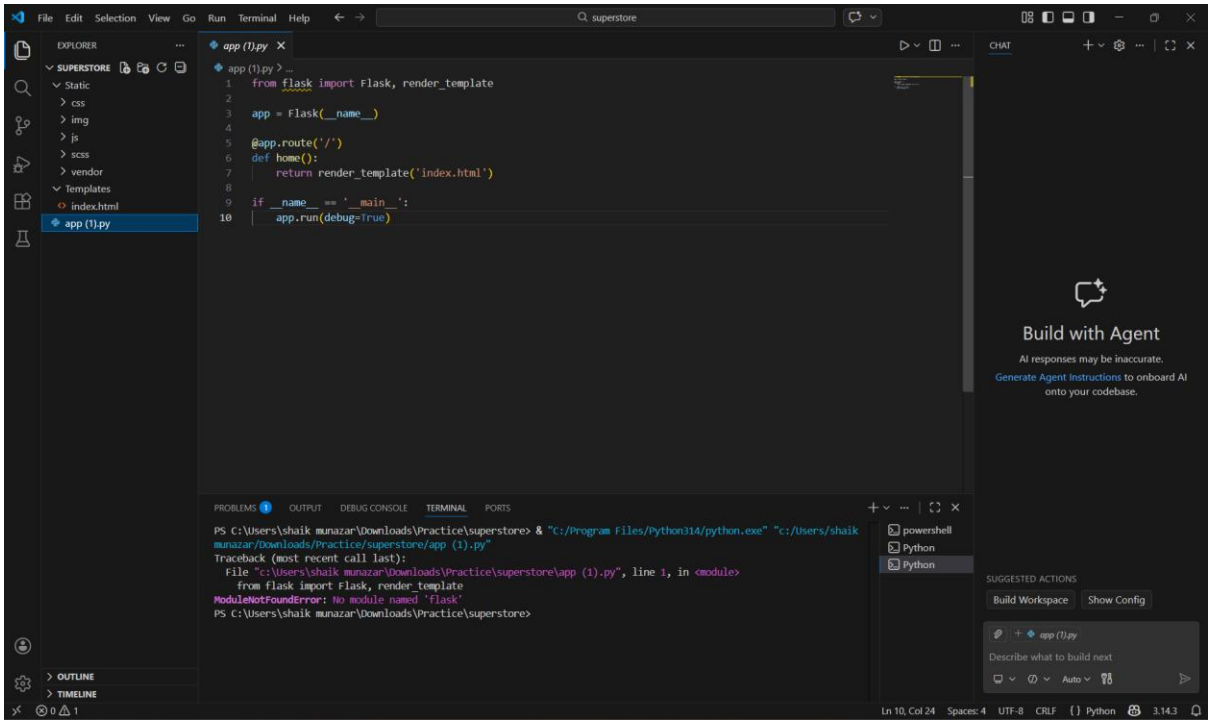
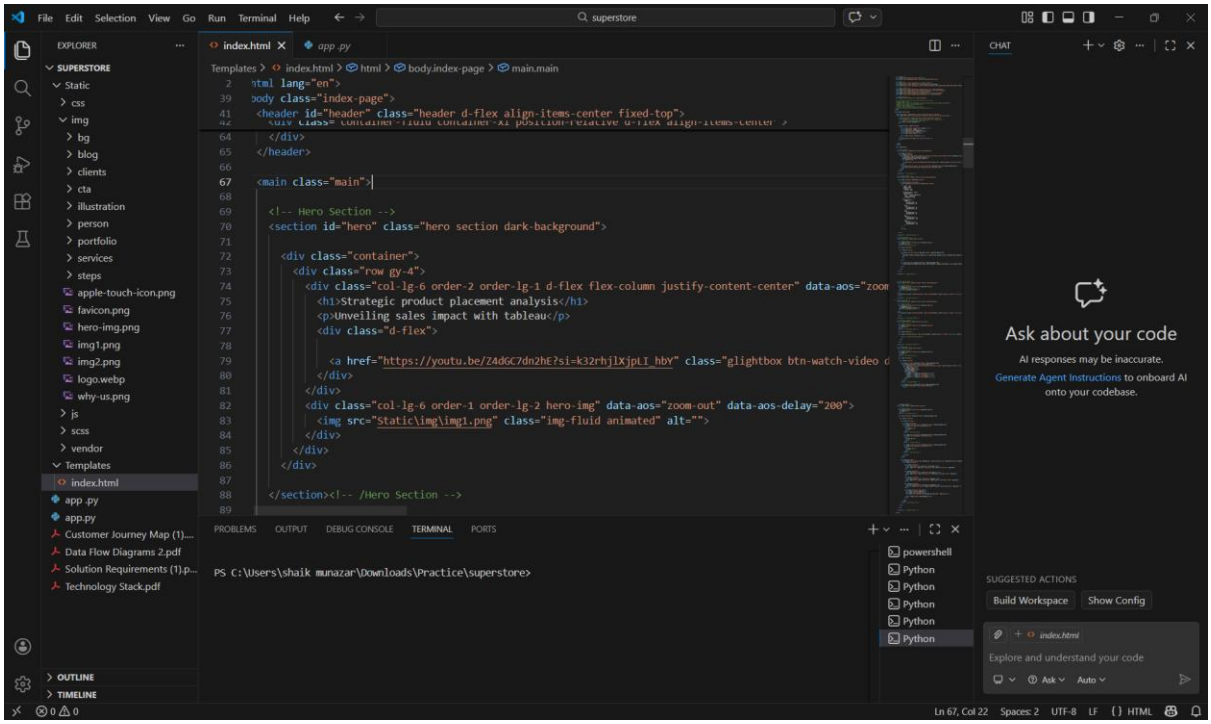
Story
Team
Contact
PRODUCT PLACEMENT ANALYSIS STORY
Category clothing has highest Average sales volume
The Product position at the front of the store as High sales volume and high foot traffic
The Product Category clothing at the front of the store ha highest avg sales
Clothing Front of Store 1,923.7
Clothing End-cap 1,747.1
Food End-cap 1,813.6
Food Aisle 1,728.2
Clothing Aisle 1,827.7
Electronics Aisle 1,760.1
Electronics End-cap 1,726.6
Food Front of Store 1,672.1
Electronics Front of Store 1,734.2
Sales Volume
173,422
213,152
View on Tableau Public
Share

VS Code Screenshots:

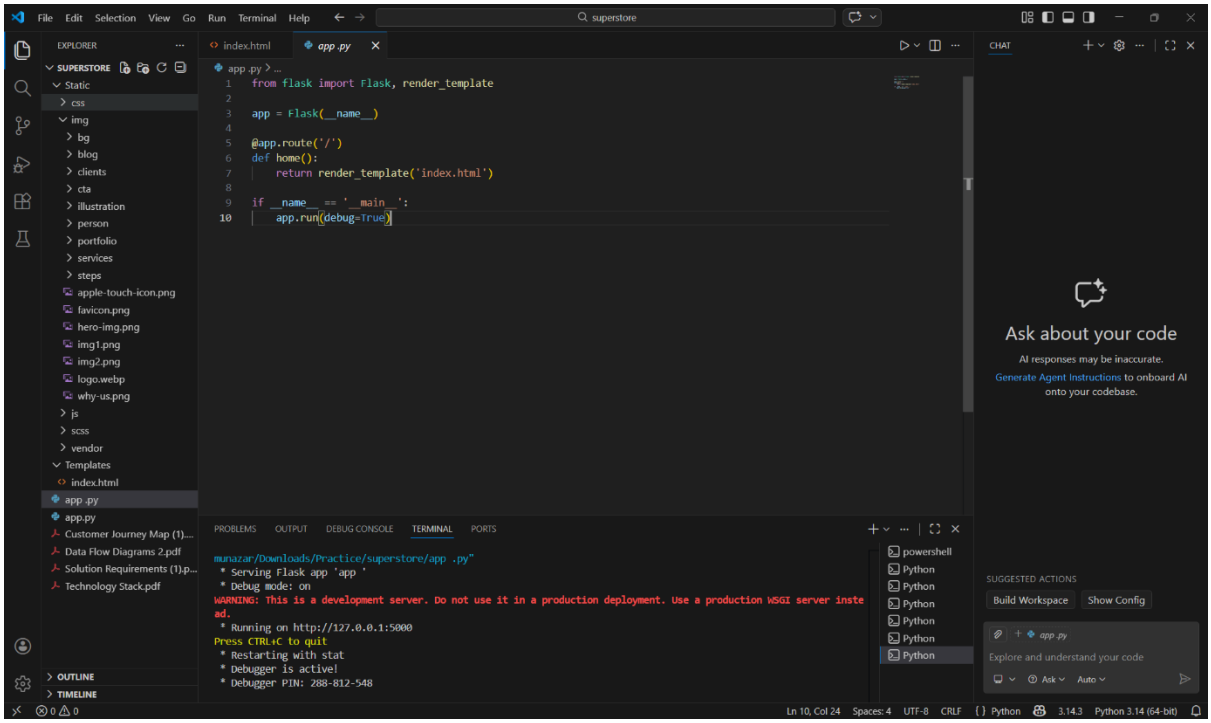
Python Flask code:



Index.html Code:



Python Flask Output:



8. ADVANTAGES & DISADVANTAGES

Advantages

• Large Dataset: Uses detailed retail sales and product placement data, making insights reliable.
• Easy to Use: Interactive filters help users analyze data without technical knowledge.
• Better Business Insights: Helps identify best product positions and categories for higher sales.
• Clear Visualization: 8 charts and story points explain placement impact clearly.

Disadvantages

• Uses Past Data: Cannot predict future sales automatically.
• Manual Updates: New data must be cleaned and updated manually.
• Limited Factors: Only available dataset fields are analyzed.
• No Prediction Model: Focuses on visualization, not machine learning forecasting.

9. CONCLUSION

The Strategic Product Placement Analysis project shows how data visualization can turn retail data into useful business insights. By using Tableau dashboards, I analyzed how product positioning, pricing, promotions, and demographics affect sales performance.

The interactive filters make the system easy to use and help businesses understand which strategies increase sales. This project proves that clear visual analysis supports better business decisions and revenue growth.

10. FUTURE SCOPE

In the future, this project can be improved by adding machine learning models to predict future sales trends.

More data like customer purchase behavior and market trends can be included for deeper analysis.

An automatic data update system can also be added so dashboards refresh in real-time.

11. APPENDIX

- Tableau Public Link: Profile - shaik.munazar | Tableau Public
- Dataset Link: https://drive.google.com/file/d/12RqLHlXBNpl5J6dWSupGXbNvYdayrs_m/view?usp=sharing
- GitHub Link:
- Project Demo: https://drive.google.com/file/d/16_p7iiP9TXvJIchO30oqPl1gVrTzUJyg/view?usp=sharing